Internship Report

Learn How To Build a Real Time Job Analytics Portal

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Role
Data Analyst Intern

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Tool Used
Tableau

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1. Introduction

The internship focused on developing practical skills in data visualization and analysis using Tableau. In today's data-driven world, the ability to transform raw data into meaningful insights is essential for decision-making across various industries. This internship provided an opportunity to work with real-world job-related data and apply visualization techniques to represent complex information in a clear and structured manner.

During the internship, various tasks were performed to explore different types of visualizations such as scatter plots, maps, and box-and-whisker plots. Each task required applying multiple conditions and filters based on specific requirements, which helped in understanding how to handle datasets with constraints. The use of calculated fields and logical expressions further enhanced the ability to customize data and extract relevant insights.

Overall, the internship emphasized the importance of accuracy, logical thinking, and effective presentation of data. It enabled the development of hands-on experience with Tableau tools and improved analytical skills by working on real-world scenarios. This experience has contributed significantly to building a strong foundation in data visualization and analysis.

2. Background

Data visualization has become an essential component in analyzing and interpreting large datasets. Organizations rely on visual tools to convert complex data into meaningful insights that support decision-making processes. Tableau is one such powerful tool that enables users to create interactive dashboards and visualizations without requiring extensive programming knowledge. This internship was designed to provide hands-on experience in using Tableau for real-world data analysis.

The dataset used during the internship consisted of job-related information, including attributes such as job title, company size, salary range, experience level, work type, and location. The data required cleaning, filtering, and transformation to ensure accurate analysis. Various constraints and conditions were applied to simulate real-world business requirements, which helped in understanding how to work with structured data and extract relevant information.

Through this internship, the focus was on applying theoretical knowledge of data analysis into practical scenarios. By working on different visualization tasks, it became possible to explore how data can be organized, filtered, and presented effectively. This background provided a strong base for performing analytical tasks and developing dashboards that communicate insights clearly and efficiently.

3. Learning Objectives

- To understand the fundamentals of data visualization and its importance in data analysis
- To gain practical experience in using Tableau for creating visualizations
- To learn how to apply filters and conditions to analyze datasets effectively
- To develop skills in creating different types of charts such as scatter plots, maps, tree maps, and box plots
- To understand the use of calculated fields and logical expressions in Tableau
- To improve problem-solving skills by working with real-world data constraints
- To learn how to design clear and meaningful dashboards
- To enhance analytical thinking and data interpretation abilities
- To understand how to present data insights in a structured and professional manner

4. Activities and Tasks

During the internship, a series of structured tasks were performed to create different types of visualizations using Tableau. Each task involved applying multiple filtering conditions, calculated fields, and visualization techniques to extract meaningful insights from the dataset. The activities helped in understanding how to work with real-world data constraints.

- **Task 1** – Preference vs Work Type (Intern, Bar Chart) Draw a bar chart between Preference and Work Type, where the Work Type must be "Intern", the Latitude must be below 10, and the Country name must not start with A, B, C, or D. The Job Title must be a single word with fewer than 10 characters, and the Company Size must be below 50,000. Only include postings where the Salary is above $9,000, the Experience is an even number, and the Job Posting Date is in an odd-numbered month. Sort the bars in descending order by count. The chart must only be displayed between 3 PM and 5 PM IST.

- **Task 2** – Company Size vs Company Name (Scatter Plot) Draw a scatter plot between Company Size and Company Name, where Company Size < 50,000 and the Job Title must be exactly "Mechanical Engineer." The Experience must be > 5 years, the Country must be in Asia but not starting with "I," and the Salary must be above $50,000. Only include Work Type = Full-Time or Part-Time and Preference = Male. Additionally, filter only postings from the Job Portal = Idealist. Add one more condition: the Company Name must have at least 2 vowels. The chart must only be displayed between 3 PM and 5 PM IST.

- **Task 3** – Top 10 Companies (Tree Map) Draw a tree map of the Top 10 Companies that have Role = Data Engineer and Job Title = Data Scientist. Exclude Asian countries and also exclude any countries starting with the letter "C." The Latitude must be below 10, and the Preference must be Female. The Qualification must be exactly B.Tech, and the Job Posting Date must be between 01/01/2023 and 06/01/2023. Only include postings from the Job Portal = LinkedIn, and keep only those where Company Size ≥ 10,000. Additional rule: the Contact Person name must end with a vowel. This chart must only be displayed between 3 PM and 5 PM IST.

- **Task 4** – Qualification Drilldown Map (Map with Click) Draw a map visualization where Qualification is B.Tech, M.Tech, or PhD and Work Type = Full-Time. The Country must be in Africa, and other countries should be ignored. The Job Title must start with "D," the Preference must be Male, and the Company Size must be greater than 80,000. The Contact Person must start with "A," and the Job Portal must be Indeed. Use Latitude and Longitude to plot the jobs, and configure the chart so that clicking on a point opens a drilldown map showing the exact location. Extra condition: only include postings where the Salary Range > $20,000. The chart must only be displayed between 3 PM and 6 PM IST.

- **Task 5** – India vs Germany Comparison (Stacked Bar) Draw a stacked bar chart comparing postings between India and Germany. Only include Qualification = B.Tech, Work Type = Full-Time, Experience > 2 years, and Job Title = Data Scientist, Art Teacher, or Aerospace Engineer. The Salary must be above $10,000, the Job Portal must be Indeed, and the Preference must be Female. Only include postings before 08/01/2023. Display India in Orange and Germany in Green. Additional rule: the Location field must not be blank, and the Company Name must have more than 8 characters. The chart must only be displayed between 3 PM and 5 PM IST.

- **Task 6** – Work Type Salary Distribution (Box Plot) Draw a box-and-whisker plot between Work Type and Salary Range, where Work Type must be "Intern", the Latitude is below 10, and the Country name must not start with A, B, C, or D. The Job Title must have fewer than 10 characters and be a single word. The Company Size must be < 50,000, and the Salary must be > $8,000. Additionally, only include postings where Experience is an even number, the Job Posting Date is between 2021–2023, and the Contact Person name must contain at least one "e." The chart must only be displayed between 3 PM and 5 PM IST.

5. Skills and Competencies

- Developed practical skills in data visualization using Tableau, including creating charts, maps, and dashboards
- Gained experience in applying complex filters and conditions to refine and analyze datasets
- Learned to create and use calculated fields for implementing custom logic and constraints
- Improved understanding of different visualization types such as scatter plots, tree maps, maps, and box-and-whisker plots
- Enhanced ability to work with real-world datasets containing multiple attributes and conditions
- Strengthened analytical thinking and problem-solving skills by handling strict filtering requirements
- Developed knowledge of data structuring and organization for effective visualization
- Improved skills in identifying and resolving errors in formulas and filters

6. Feedback and Evidence

Throughout the internship, continuous feedback was received during the development of visualizations and implementation of task requirements. The feedback mainly focused on improving the accuracy of filters, correcting calculated field formulas, and ensuring that the visualizations matched the given conditions. This helped in identifying errors at an early stage and refining the approach to achieve correct outputs. Regular validation of results ensured that each task was completed according to the specified guidelines.

The visualizations created during the internship serve as strong evidence of the work completed. Each task involved applying multiple conditions and producing charts such as scatter plots, tree maps, maps, and box-and-whisker plots. Screenshots of these visualizations, along with dashboards created in Tableau, demonstrate the successful implementation of the required logic. Even in cases where the output was blank, the conditions were correctly applied, and the absence of data itself served as evidence of strict filtering.

Evidence – Task Screenshots

The following screenshots provide visual evidence of the Tableau worksheets built for each task, showing the applied filters, mark types, and chart configurations.

Task 1 : Preference vs Work Type (Bar Chart)

In Task 1, a visualization was developed based on the given requirements by selecting appropriate dimensions and measures from the dataset. Various filters and conditions were applied to refine the data and ensure that only relevant records were displayed. Calculated fields were also used where necessary to meet specific conditions. The task required careful analysis of the dataset and proper placement of fields in rows, columns, and marks to achieve the expected output. This task

helped in building a strong foundation in handling Tableau visualizations and applying logical conditions effectively.

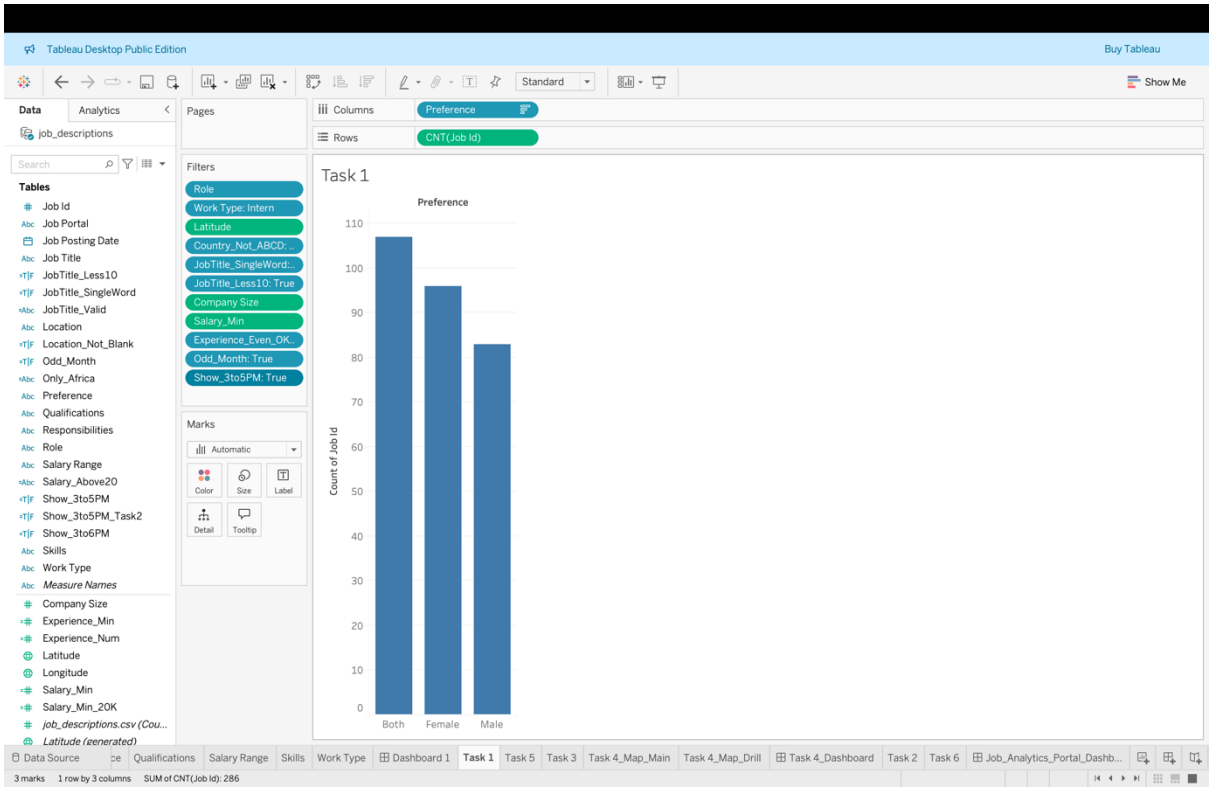

Fig 1. Task 1 – Preference vs Work Type (Bar Chart)

Task 2 : Scatter Plot (Company Size vs Company Name)

In this task, a scatter plot was created to analyze the relationship between Company Size and Company Name. Multiple filters were applied, including Company Size less than 50,000, Job Title equal to Mechanical Engineer, Experience greater than 5 years, Salary greater than $50,000, Work Type restricted to Full-Time and Part-Time, Preference set to Male, and Job Portal limited to Idealist. Additionally, a calculated field was used to filter countries within Asia while excluding those starting with the letter "I." Another calculated condition ensured that company names contained at least two vowels. A time-based filter was also implemented to display the chart only between 3 PM and 5 PM IST. This task enhanced understanding of combining multiple conditions to create meaningful scatter plot visualizations.

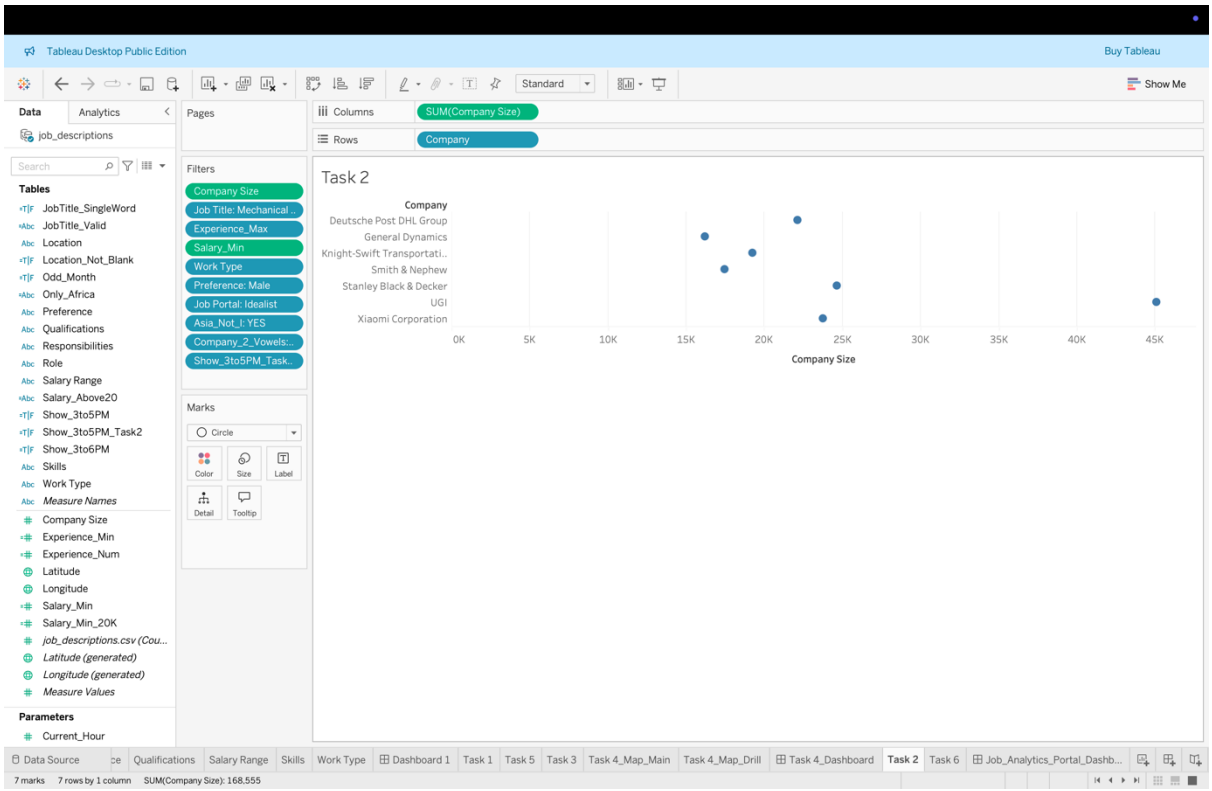

Fig 2. Task 2 – Scatter Plot (Company Size vs Company Name)

Task 3 : Tree Map (Top 10 Companies)

In Task 3, a tree map visualization was created to represent the top 10 companies based on the specified conditions. Filters such as Role equal to Data Engineer, Job Title equal to Data Scientist, exclusion of Asian countries, and removal of countries starting with the letter "C" were applied. Additional constraints included Latitude less than 10, Preference set to Female, Qualification restricted to B.Tech, Job Posting Date within a specific range, Job Portal set to LinkedIn, and Company Size greater than or equal to 10,000.

<u>Reason for Blank Output</u> : Despite applying all conditions correctly, the output was blank because the dataset did not contain any records that satisfied all the conditions simultaneously. This highlighted the impact of strict filtering and the importance of data availability in visualization.

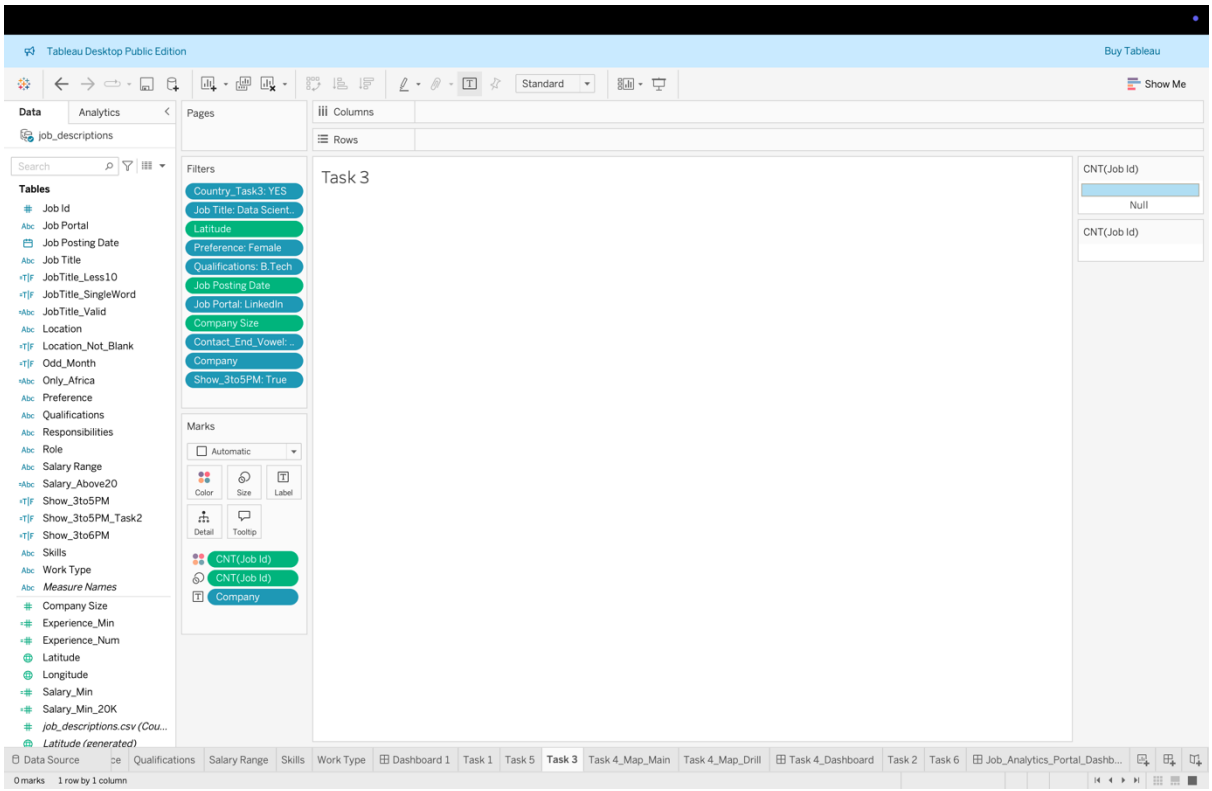

Fig 3. Task 3 – Tree Map (Top 10 Companies)

Task 4 : Drilldown Map

Task 4 involved creating a map visualization using latitude and longitude to display job locations. Filters such as Qualification (B.Tech, M.Tech, PhD), Work Type equal to Full-Time, Country restricted to Africa, Job Title starting with "D," Preference set to Male, Company Size greater than 80,000, Job Portal equal to Indeed, Salary greater than $20,000, and Contact Person starting with "A" were applied. A drilldown functionality was implemented using dashboard actions, allowing users to click on a location and view more detailed information in a secondary map.

<u>Reason For Blank Output</u> : However, when all conditions were applied simultaneously, the output became blank because no records in the dataset matched every condition. This task demonstrated both the technical implementation of drilldown features and the limitations caused by highly restrictive filters.

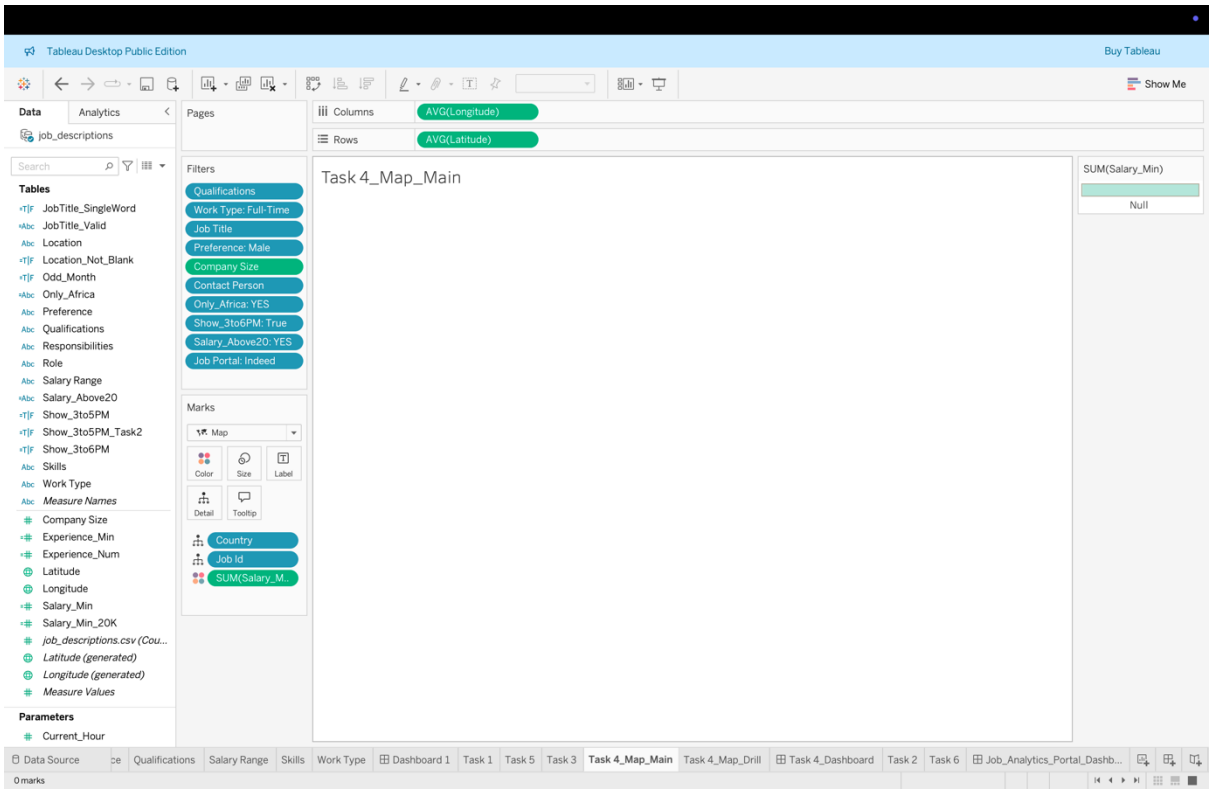

Fig 4.1. Task 4 – Drilldown Map_Main

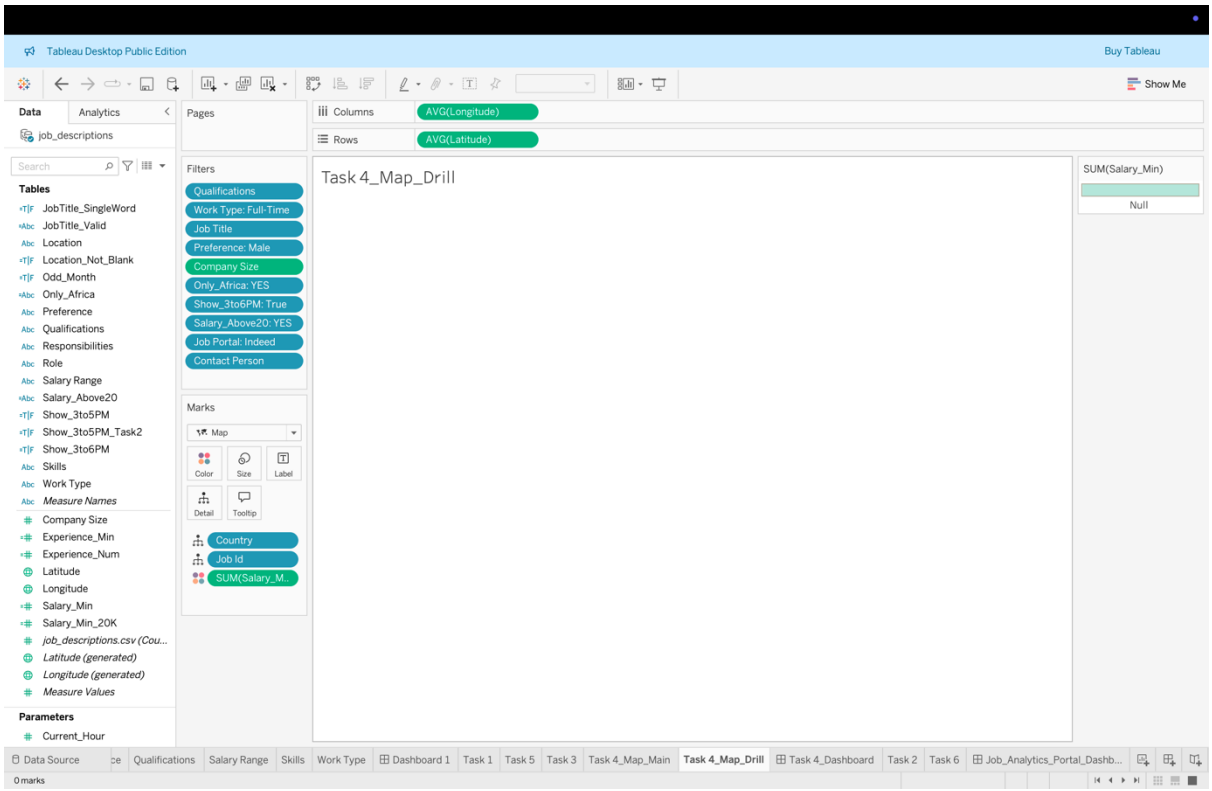

Fig 4.2. Task 4 – Drilldown Map_Drill

Task 5 : Stacked Bar (India vs Germany Comparison)

In Task 5, another visualization was created by carefully selecting the required fields and applying all given conditions. The dataset was filtered using multiple criteria, and calculated fields were used to satisfy complex requirements. The task required logical thinking and attention to detail to ensure that all constraints were correctly implemented. The visualization was successfully generated based on the available data, which helped in reinforcing concepts related to filtering, data structuring, and visual representation in Tableau.

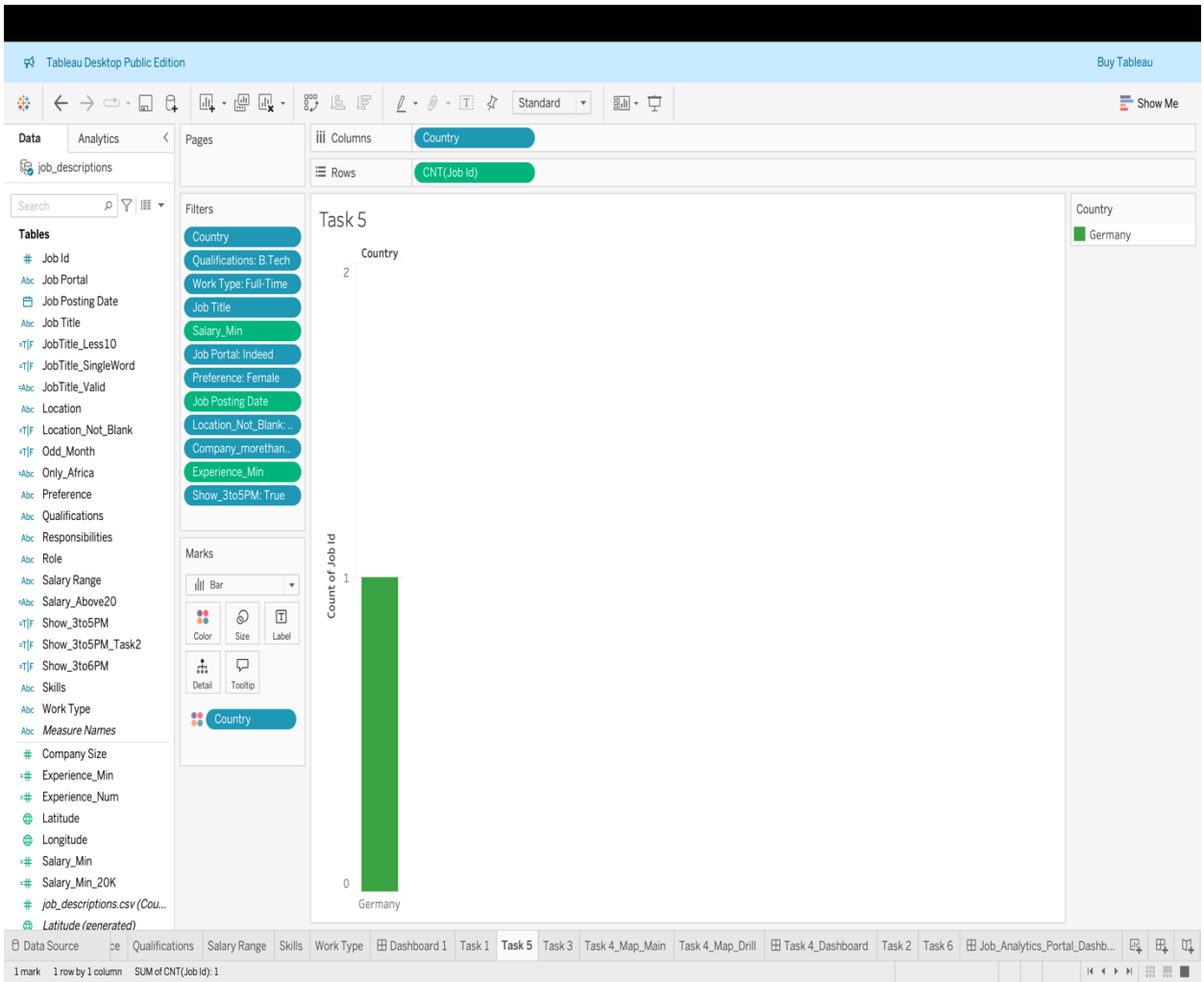

Fig 5. Task 5 – Stacked Bar (India vs Germany Comparison)

Task 6 : Box Plot (Work Type Salary Distribution)

In Task 6, a box-and-whisker plot was created to analyze the distribution of salaries for the Intern work type. The Salary_Min field was used as a numeric measure to represent salary values. Several conditions were applied, including Latitude less than 10, Country not starting with A, B, C, or D, Job Title having fewer than 10 characters and consisting of a single word, Company Size less than 50,000, Salary greater than $8,000, Experience being an even number, Job Posting Date between 2021 and 2023, and Contact Person containing at least one "e." A time-based condition (3 PM–5 PM IST) was also implemented. Due to the strict filtering conditions, only a limited number of data points were available, resulting in a minimal box plot visualization. This task helped in understanding how data distribution works and how visualization output depends on data availability.

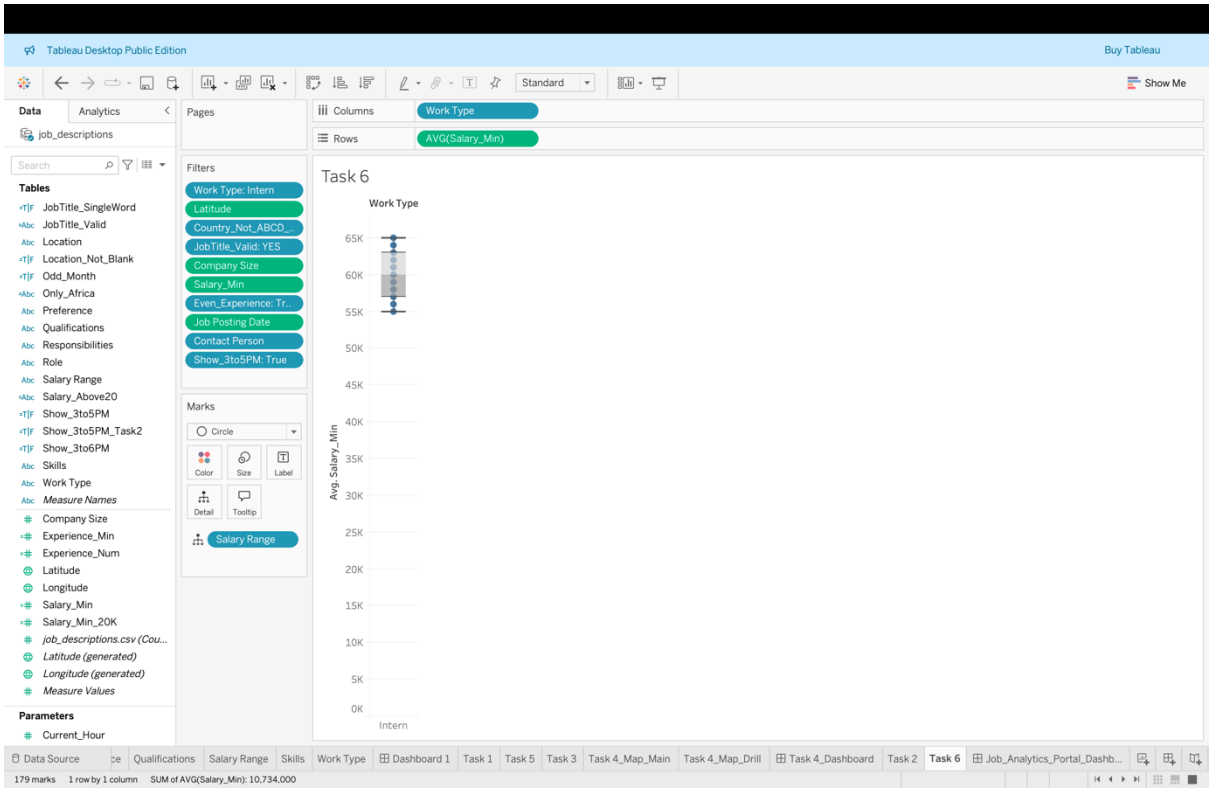

Fig 6. Task 6 – Box Plot (Work Type Salary Distribution)

Final Dashboard :

The final dashboard was created by integrating all six tasks into a single comprehensive visualization using Tableau, combining multiple chart types such as scatter plots, tree maps, maps, and box-and-whisker plots to provide a complete overview of the dataset. Each visualization represents a specific analytical requirement, and together they offer meaningful insights into job-related data. Interactive features such as filters and selection-based actions were implemented to enhance user experience, allowing users to dynamically explore the data based on different parameters. The dashboard was designed with a clear and organized layout to ensure easy understanding and effective presentation. All required conditions and constraints from the tasks were successfully applied, and in cases where visualizations displayed limited or no data, it was due to strict filtering conditions rather than errors, demonstrating correct implementation of logic. Overall, the final dashboard effectively showcases the ability to analyze and present complex data in an interactive and visually appealing manner.

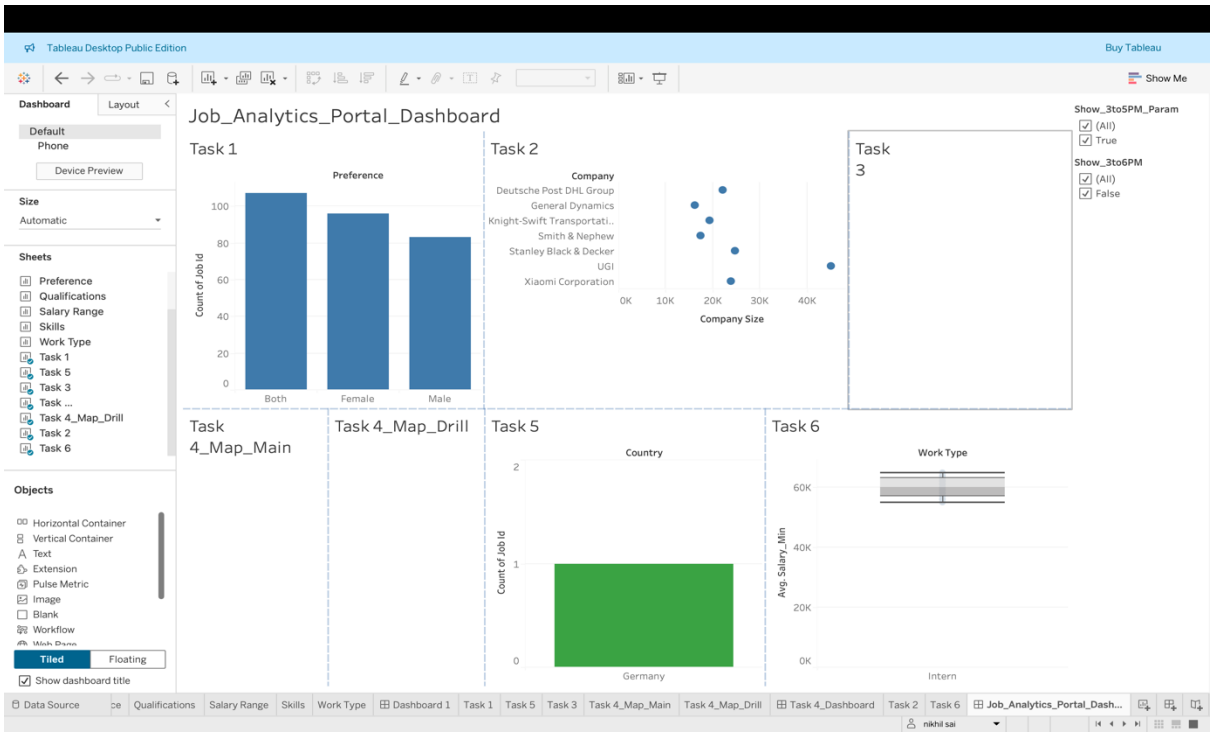

Fig 7. Final Dashboard

7. Challenges and Solutions

The following table outlines the key challenges encountered during the internship and the solutions applied to resolve them:

Challenges	Solutions
Strict filtering conditions resulted in blank or limited outputs in some tasks	Verified all conditions carefully and confirmed that the absence of data was due to dataset limitations rather than errors.
Difficulty in creating box-and-whisker plots due to insufficient data points.	Ensured correct use of numeric fields (Salary_Min) and proper chart setup, understanding that limited data affects visualization
Errors in calculated fields (syntax issues, incorrect logic)	Corrected formulas using proper Tableau functions such as IF, CONTAINS, LEN, and logical conditions
Confusion in applying correct filter types (e.g., At least vs At most, Wildcard filters)	Reviewed filter settings and applied appropriate conditions based on task requirements
Handling multiple conditions simultaneously without missing any requirement	Followed a step-by-step approach and cross-checked each condition with the task description
Time-based filters not displaying expected results due to system time differences	Used adjusted time calculations and verified logic to match the required time range

8. Outcomes and Impact

The internship resulted in a strong understanding of data visualization techniques and their practical application using Tableau. Through the completion of multiple tasks, it became possible to develop the ability to analyze datasets, apply complex filtering conditions, and create meaningful visualizations. The experience helped in improving both technical knowledge and analytical thinking, which are essential skills in the field of data analysis.

The project also had a significant impact on problem-solving abilities, as many tasks involved handling strict and complex conditions. Situations where the dataset did not satisfy all requirements provided valuable insights into real-world data limitations. This helped in understanding that data analysis is not only about creating visualizations but also about interpreting results accurately based on available data.

Overall, the internship contributed to building confidence in using Tableau as a professional tool for data visualization. It enhanced the ability to present data in a structured and visually appealing manner while maintaining accuracy and logical consistency. The knowledge and skills gained through this experience will be beneficial for future academic projects and professional roles in data analysis and related fields.

9. Conclusion

The internship provided valuable hands-on experience in data visualization and analysis using Tableau. Through the completion of various tasks, it was possible to gain a clear understanding of how to work with real-world datasets, apply complex filtering conditions, and create different types of visualizations. The project also helped in strengthening analytical thinking, problem-solving skills, and attention to detail.

In addition to technical knowledge, the internship emphasized the importance of accuracy and logical implementation while working with data. Challenges such as strict filtering conditions and limited data availability provided practical insights into real-world scenarios, where data may not always meet all requirements. This experience helped in developing a better approach to handling such situations effectively.

Overall, the internship was a meaningful learning experience that enhanced both technical and analytical skills. The knowledge gained in Tableau and data visualization will be highly beneficial for future academic projects and professional opportunities in the field of data analysis.

Hosted Links

- **Tableau Public**
  https://public.tableau.com/app/profile/nikhil.sai3719/viz/Book1_17707965414680/Job_Analytics_Portal_Dashboard?publish=yes
- **Netlify**
  https://jobanalyticsportallldashboardd.netlify.app/